SAISHUBHAM LAISETTI

cs24mtech12015@iith.ac.in | SAISHUBHAMLAISETTI@GMAIL.COM | +91 6300654096 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.63
Sri Aakash Junior College	Class XII (Telangana State Board of Intermediate Education)	2020	88.30%
KNARAYANA HIGH SCHOOL	Class X (Board Of Secondary Education Andhra Pradesh)	2018	8.30%

WORK EXPERIENCE

Scalable Algorithms for Large-Scale Data (ScaleX) Lab, IIT Hyderabad

- **Research in scalable algorithms** for large, high-dimensional data, **compressing big datasets into compact "sketches" with provable accuracy** guarantees.
- Working on tensor sketching, randomized linear algebra, sublinear algorithms **for AI and ML and signal processing applications.**

Web Development and Designing Intern at **Oasis Infobyte**

- **Developed responsive web interfaces** by improving usability across desktop and mobile platforms.

PROJECTS

Real Time application for JLPT N5 Trainer -- Interactive Japanese Learning Platform | [Live Demo]

- Improved learning retention using **Leitner 5-box spaced repetition**, prioritizing missed vocabulary for adaptive review.
- **Streamlined JLPT preparation** through **adaptive quizzes and 12-week study plans**, enabling **structured daily practice**.
- Enhanced learning continuity with **IndexedDB persistence**, retaining **progress, streaks,** and exam history.
- Tech Stack : JavaScript, HTML/CSS, **IndexedDB, Web Speech API**

Real Time CIP Day 2026 Live Scoring Platform -- Tournament Management Platform | [Live Demo]

- Built a live bracket and results site for a campus sports day with 8 events and 44 participants.
- **Used GitHub Pages** with the **GitHub REST API** as a **serverless backend**, with **token-based organiser-only editing.**
- Supported knockout, team and single-event formats with **organiser-controlled draws and scheduling**.
- Tech Stack : JavaScript, HTML/CSS, GitHub REST API, GitHub Pages

Machine Learning Approaches for Gap Acceptance Modelling

- Modeled driver Gap Acceptance using **CatBoost** and **Logistic Regression**.
- Achieved **superior Recall (0.98) with CatBoost**, and used **SHAP** analysis for predictor confirmation.

Quora Question Pair Similarity | NLP, Machine Learning

- **Engineered** a machine learning pipeline to detect duplicate Quora questions by applying **text preprocessing, TF-IDF weighted Word2Vec embeddings, and feature engineering** resulting in improved semantic similarity detection and robust model performance.
- **Implemented** classification models including **Logistic Regression, Linear SVM, and XGBoost** using train-test splits and evaluation metrics (Accuracy, F1, ROC-AUC) achieving reliable baseline comparisons for text similarity tasks.
- **Visualized** high-dimensional feature spaces through **EDA and t-SNE techniques** enabling deeper insights into data

distribution and feature relevance for question pair similarity.

PUBLICATIONS

Drowsiness Detection System: Stay Alert Stay Alive [Link]

SKILLS

- **Agentic AI & AI Development**: Claude Code, OpenAI Codex, Google Antigravity, **AI Agents, Agentic Workflows**, LLM Tool Use, **RAG, MCP**, Prompt Engineering, **AI-Assisted** Software Development, Semantic Search, Vector Databases.
- **AI & ML:** PyTorch, scikit-learn, Deep Learning, Computer Vision, Transformers, NLP, Generative AI, Model Evaluation, SHAP.
- **Programming & Development:** Python, C++, Data Structures & Algorithms, Object-Oriented Programming. **(500+) Problems Solved**
- **Data Science & Signal Processing:** NumPy, Pandas, Feature Engineering, Wavelet Transforms, Statistical Analysis
- **Tools & Platforms:** Linux, Git, GitHub, Docker, Hydra, Weights & Biases (W&B)

POSITIONS OF RESPONSIBILITY

- **Hostel Representative** - Ensured that no student should suffer or face any difficulty throughout the IIT Hyderabad Journey.
- **OCS Placement Coordinator** Strengthened campus placements by coordinating recruiter-student communication.
- **Anchor for E-Cell Milan 2024** managing **live hosting** and audience interaction and even managing**.**
- **Mess Coordinator** Improved **dining standards for 5,700+ students** by monitoring food quality, hygiene, and feedback.
- Expanded **educational outreach to 7+ government schools** through **Prayas**, providing **career guidance** to rural students.
- **Organized 12+ cultural events annually** through **SPIC MACAY**, coordinating **monthly programs** promoting Indian arts and heritage.

EXTRACURRICULAR ACTIVITIES & ACHIEVEMENTS

- **1st Place - Japan Sports Day** for **2 consecutive years**, demonstrating sustained **cultural and athletic engagement**.
- **Inter-IIT Sports Meet:** Set the **best Men's 800m & 1500m records** at **IIT Hyderabad to date**.
- **Gold Medalist - Men's 800m**, Milan 2025 Inter-Hostel General Championship, IIT Hyderabad.
- **Full Marathon Finisher:** Completed the **42.195 km** NMDC Hyderabad Marathon 2025.
- **Gold Medalist -** CSE Day **Badminton Tournament** 2026.
- **Gold Medalist (2026)** and **Runner-Up (2025)** in Cricket at the CSE Day **Cricket Tournament**.
- **Silver Medalist - Men's 400m**, recording a **1:00.65 personal best** at the **IITH Athletics League 2025-26**.
- **Top 30 Contestant &** Highest Prize-Money Winner in **Meelo Evaru Genius**, competing among **1,000+ students**.

RELEVANT COURSES

Spoken Japanese Basics | Linear Optimization | Fraud Analytics | Advanced **Data Structures and Algorithms |** Practical Malware Analysis | **Foundations of Machine Learning |** Network Security | **Tensor**: Techniques, Algorithms & Applications, Data Mining | Representation Learning

INTERESTS

- **Research**, AI & ML, Athletics, **Cooking**, Cricket, Badminton, **Trekking**, Reading Books, Formula 1.

LANGUAGE LEARNING

- **Learning Japanese** (spoken & written) through **self-study and coursework** in Spoken Japanese Basics.
- Preparing for **JLPT** (2027).